Daisy Seed 3 Pedal Dev Kit Rev3 — Reference Design Review

Date: 2026-10-12 · **Inputs:** Electrosmith's published Rev3 KiCad project and production Gerbers (2025-12-10) from `github.com/daisyaudio/ES-Seed3-DevKit-Pedal`, the Rev4 BOM from the Daisy CDN (`reference/electrosmith/seed3_pedal_devkit/`) · **Scope:** Passes 1–6, 78 ledger entries (`FINDINGS.md`): 4 violations, 5 zero-margin, 43 passes, the rest judgment · **Tooling:** KiCad 10.0.7; Pass 1 showed the committed production Gerbers are this exact project state.

The short version

This is the board Trey's power entry was copied from, and the copy is exact. The Pedal kit's chain is the Seed 3 datasheet's Figure 1.1: three beads, a single series Schottky, one more bead, a 100 µF split node, then two 3.3 Ω / 100 µF / 3.3 Ω / 100 µF π-filters, one to Seed VIN and one to the analog 9 V rail. Trey's FB3/FB4/FB5 → D6 → FB6 → C19 → R29/R30 and R31/R32 chains are the same components in the same order. The differences are what Trey added and what he loads it with.

The bead-chain answer (7 October items 2 and 9)

	Pedal kit Rev3	**WavetableController rev 3**
Bead part	Chilisin PBY160808T-601Y-N, 600 Ω @100 MHz, **1 A**, 200 mΩ (PD-P2-01)	Murata C14709, 120 Ω, 2 A, 50 mΩ
Reverse-polarity scheme	the single series Schottky D5 is the whole scheme	4-diode bridge D2–D5 **plus** the series diode D6
Series diode current	40–130 mW, ≤65 % of rating at 0.3 A	≈0.2 W, near rating (R3-P2-06)
Second π branch load	17 mA (op-amps + LDO)	0.22–0.30 A (OLED buck + LEDs)
Bulk behind the beads	200 µF, no inrush limit (PD-P2-03)	200 µF, no inrush limit (R3-P2-05)
π resistors	3.3 Ω 1206, 0.75 W	3.3 Ω 2512, 1 W

So: the 7 October review's "1 A bead" claim is confirmed; D6 is the copy and the bridge is the addition; the inrush exposure is inherited from the reference (the kit's 200 mΩ beads actually limit it four times better than Trey's 50 mΩ ones); and the reason Trey's "+9V" rail sags to 5.8 V while the kit's analog rail sits at 8.2–8.5 V is the 15× heavier load on that branch, not the circuit. The recommendation to feed the OLED buck from the C19 node, or delete the second π stage, stands.

Tier 1 — Fab-rule violations in the reference (marginal; built this way)

- **Ground pours 0.025 mm from the routed edge** (PD-P1-06), same as the Eurorack kit, vs JLCPCB 0.2.
- **Toggle SW1 pads 0.161 mm ring** on three pads (PD-P1-04); footswitch pads at 0.254 on the 0.25 recommendation.
- **USB-C pegs 0.175 mm from GND pads** (PD-P1-07). KiCad 9.0.9 misses these two `hole_clearance` errors; 10.0.7 reports them.
- **Silk:** 62 strokes and 12 heights under JLC minimums, 39 of them user-facing legends (PD-P6-02).

Tier 2 — Zero-margin and reference habits

- **Expression input on 3.3 V-only pin 28** fed from a 5 V MCP6002 follower through a 10k/20k divider: 3.33 V nominal, 3.36–3.38 V worst case, the datasheet's own Figure 4.5 (PD-P3-07, PD-P2-11).
- **+3V3_A has no capacitor at all** (PD-P2-06); +3V3 has one 100 nF at the opto, 209 mm routed from the Seed; SD socket 239 mm from any cap (PD-P2-05). Same family of omission as the Pod, the Eurorack kit and rev 3.

- **MIDI IN has no anti-parallel diode** across the opto LED (PD-P3-09); the Pod, the Eurorack kit and rev 3 all have one. THRU makes the H11L1 sink 21 mA (PD-P2-16).
- **Nine unfiltered switch inputs**, footswitch runs 116–148 mm on internal pull-ups (PD-P5-07).
- **4 GND vias only**, 38 of 65 SMD GND pads 20–35 mm from any layer link (PD-P1-10).
- **Documentation drift:** README switch refdes are the Rev1 names, sheet files are still named Rev1, the README links a Rev4 BOM that adds a D6 the Rev3 board doesn't have, and the Rev3 repo's only BOM has no MPN/DNP/assembly columns (PD-P4-02, PD-P6-06, PD-P2-18). The production BOM merges two footprints on one 10 µF line (PD-P4-01).

Tier 3 — Things the kit does better than rev 3

- **Card detect is wired** to pin 12 with libDaisy support, the only board in the set that can see a card (PD-P3-05).
- **Audio is buffered** both ways (OPA1652 on +9V_A, 1 MΩ AC-coupled inputs, 100 Ω outputs) and no digital trace runs within 1 mm of any analog trace (PD-P5-05), the cleanest of the three references; MIDI lines stay 180 mm from the audio.
- **Every part is on a BOM with an Assembly column**, THT included.
- **Four 3.5 mm mounting holes** at 5 mm from every edge (PD-P6-07).

What passed (verified, cited)

Schematic ↔ board 150/150, 0 pad diffs; 0 unconnected; min copper 0.20 mm; 0 clearance errors at JLC minimums; production Gerbers = project; paste ↔ BOM 286/286; 39/40 Seed pins on nets, firmware map 30/30; SD guideline met; USB pair 33 ps skew, 0 vias; power per datasheet Figure 1.1 with DSY_VIN 6.4–7.9 V; LP2985 and OPA1652 decoupled per datasheet; pots direct to ADC0–5.

Caveats

Enclosure drawing not published, so hole positions are reported, not checked. Microchip, Kingbright, H11L1 and Chilisin datasheets unreachable; the bead rating comes from JLCPCB's part page and the Rev4 CDN BOM. STM32 absolute maximum ratings from memory, flagged.